

Week 4 Assignment: JavaScript Fundamentals – Part 2

Duration: 1 Week (7 Days) | Marks: 100 | Due: 11 Oct

Submission Format: ZIP Folder + Google Drive Link

■ Objective

The objective of this assignment is to build upon the JavaScript fundamentals learned in Week 3 and understand how Functions, Arrays, Objects, DOM Manipulation, and Events are used to create interactive web pages. Students will apply these concepts to develop dynamic and interactive JavaScript-based webpages.

■ Instructions

- Complete all the tasks in sequence.
- Use the HTML and CSS concepts learned in previous weeks.
- Write clean, properly structured JavaScript code.
- Use an external JavaScript file wherever possible.
- Test all functionalities before submission.
- Submit all HTML, CSS, and JavaScript files in a single ZIP folder.

■ Task 1: JavaScript Functions (20 Marks)

Create JavaScript programs demonstrating the use of functions.

Include:

- Function Declaration
- Function Parameters
- Function Arguments
- Return Values
- Multiple Functions
- Arrow Functions

Create meaningful examples such as calculating marks, calculating total price, generating a greeting, or calculating a result.

■ Task 2: Arrays (20 Marks)

Create JavaScript programs using Arrays.

Demonstrate:

- Creating Arrays
- Accessing Array Elements
- Adding Elements
- Removing Elements
- Updating Elements
- Iterating through Arrays
- Basic Array Methods

Use a practical dataset such as student names, products, courses, or employee records.

■ Task 3: Objects (15 Marks)

Create and work with JavaScript Objects.

Your program should demonstrate:

- Creating Objects
- Object Properties
- Object Values
- Accessing Properties
- Updating Properties
- Adding New Properties
- Removing Properties

Create at least 3 meaningful objects based on a practical scenario.

■ Task 4: DOM Manipulation (20 Marks)

Create an HTML webpage and use JavaScript to dynamically interact with its elements.

Demonstrate:

- Selecting HTML Elements
- Changing Text Content
- Changing HTML Content
- Changing CSS Styles
- Adding Elements
- Removing Elements
- Updating webpage content dynamically

■ Task 5: Events (15 Marks)

Add JavaScript Events to your webpage.

Implement events such as:

- click
- input
- change
- submit
- Mouse/Keyboard Events

Create meaningful interactions where the webpage responds to user actions.

■ Task 6: Mini JavaScript Project (10 Marks)

Develop a small interactive JavaScript project by combining the concepts covered in Week 4.

Your project should use:

- Functions

- Arrays
- Objects
- DOM Manipulation
- Events

Example Projects:

- To-Do List
- Student Record Manager
- Simple Quiz Application
- Expense Tracker
- Notes Application
- Product List

The project should be functional and user-friendly.

■ Submission Guidelines

Submit all HTML, CSS, and JavaScript files in a single ZIP folder.

File Name Format: WD_Week4_YourName.zip

Example: WD_Week4_RahulSharma.zip

Upload the ZIP folder to Google Drive, enable "Anyone with the link can view" access, and submit the Google Drive link on the internship portal.

■ Evaluation Criteria (100 Marks)

Task	Marks
Task 1 – JavaScript Functions	20
Task 2 – Arrays	20
Task 3 – Objects	15
Task 4 – DOM Manipulation	20
Task 5 – Events	15
Task 6 – Mini JavaScript Project	10
Total	100

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